The Living Cell

A short guide to the smallest unit of life. Chapter 1

> **In one sentence:** every living thing is made of cells, and every cell comes from another cell.

1. What a cell is

A cell is a tiny, self-contained unit that takes in food, turns it into energy, responds to its surroundings and makes copies of itself. Some living things, such as bacteria, are a single cell. A human body is made of around thirty trillion cells of more than two hundred kinds, from long nerve cells to flat skin cells.

Cells are small because of a simple rule of geometry. As a cell grows, its volume increases faster than its surface. The surface is where food and oxygen come in and waste goes out, so a very large cell could not feed its own middle. Staying small keeps every part of the cell close to its surface.

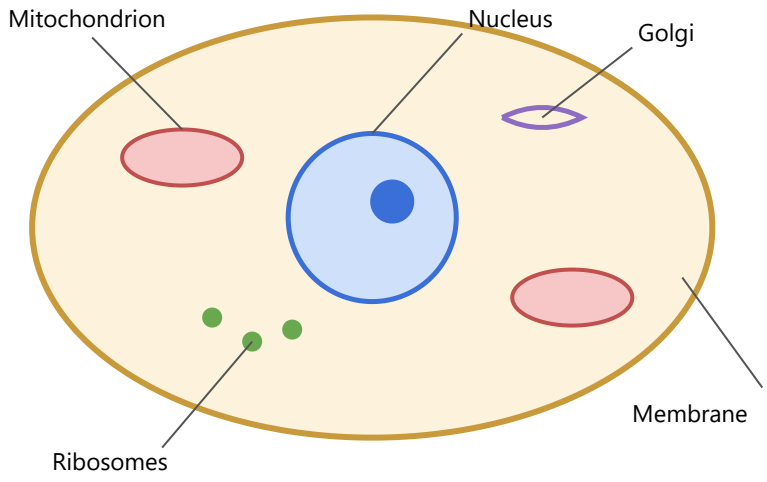


Figure 1. A typical animal cell and its main parts.

2. The cell membrane

Every cell is wrapped in a membrane, a thin double layer of fat-like molecules with proteins set into it. The membrane decides what enters and leaves. Small molecules such as oxygen slip straight through; larger ones such as glucose need protein doors called channels and carriers.

3. Energy: the mitochondria

Mitochondria are the power stations of the cell. They break down sugar using oxygen and store the energy released in a molecule called ATP. A muscle cell, which needs a great deal of energy, can hold thousands of mitochondria.

> **Remember:** respiration in the mitochondria turns glucose and oxygen into carbon dioxide, water and ATP.

4. Instructions: the nucleus

The nucleus holds the cell's DNA, the instructions for building every protein the cell needs. Copies of these instructions, written in a molecule called messenger RNA, leave the nucleus and are read by ribosomes, which join amino acids into proteins in the right order.

5. Plant and animal cells compared

Part	Animal cell	Plant cell	What it does
Cell membrane	Yes	Yes	Controls what enters and leaves
Nucleus	Yes	Yes	Holds the DNA
Mitochondria	Yes	Yes	Release energy as ATP
Cell wall	No	Yes	Gives shape and strength
Chloroplasts	No	Yes	Make sugar from light
Large vacuole	No	Yes	Stores water, keeps the cell firm

6. How cells divide

A cell makes two cells by copying its DNA and then splitting in two, a process called mitosis. Each new cell receives a full, identical set of instructions. This is how a body grows and how it repairs a cut.

7. Questions to test yourself

Q: Why are cells so small?

Q: Which part of the cell releases energy, and in what form?

Q: Name three parts found in plant cells but not in animal cells.

Q: What does mitosis produce?